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What Is Transitive Dependency in a Database

Avoid transitive dependencies to help ensure normalization

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A transitive dependency in a database is an indirect relationship between values in the same table that causes a functional dependency. To achieve the normalization standard of Third Normal Form (3NF), you must eliminate any transitive dependency.

By its nature, a transitive dependency requires three or more attributes (or database columns) that have a functional dependency between them, meaning that Column A in a table relies on Column B through an intermediate Column C. Let's see how this might work.

Transitive Dependency Example

AUTHORS

Author_ID	Author	Book	Author_Nationality
Auth_001	Orson Scott Card	Ender's Game	United States

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	Card	Mind	
Auth_002	Margaret Atwood	The Handmaid's Tale	Canada

In the AUTHORS example above:

Book → Author: Here, the *Book* attribute determines the *Author* attribute. If you know the book name, you can learn the author's name. However, *Author* doesn't determine *Book*, because an author can write multiple books. For example, just because we know the author's name is *Orson Scott Card*, we still don't know the book name.

Author → Author_Nationality: Likewise, the *Author* attribute determines the *Author_Nationality*, but not the other way around—just because we know the author's nationality doesn't mean we can determine the author.

But this table introduces a transitive dependency:

Book → Author_Nationality: If we know the book name, we can determine the author's nationality via the Author column.

Avoiding Transitive Dependencies

To ensure Third Normal Form, let's remove the transitive dependency.

We can start by removing the Book column from the Authors table and creating a separate Books table:

BOOKS

Book_ID	Book	Author_ID
Book_001	Ender's Game	Auth_001
Book_001	Children of the Mind	Auth_001
Book_002	The Handmaid's Tale	Auth_002

AUTHORS

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Auth_002	Margaret Atwood	Canada

Did this fix it? Let's examine our dependencies now:

BOOKS table:

Book_ID → **Book**: The *Book* depends on the *Book_ID*.

No other dependencies in this table exist, so we are okay. Note that the foreign key, *Author_ID*, links this table to the AUTHORS table through its primary key, *Author_ID*. We have created a relationship to avoid a transitive dependency, a key design of relational databases.

AUTHORS table:

Author_ID → **Author**: The *Author* depends on the *Author_ID*.

Author → **Author_Nationality**: The nationality can be determined by the author.

Author_ID → **Author_Nationality**: The nationality can be determined from the *Author_ID* through the *Author* attribute. We still have a transitive dependency.

We need to add a third table to normalize this data:

COUNTRIES

Country_ID	Country
Coun_001	United States
Coun_002	Canada

AUTHORS

Author_ID	Author	Country_ID
Auth_001	Orson Scott Card	Coun_001
Auth_002	Margaret Atwood	Coun_002

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The BOOK table's foreign key, *Author_ID*, links a book to an author in the AUTHORS table.

The AUTHORS table's foreign key, *Country_ID*, links an author to a country in the COUNTRIES table.

The COUNTRIES table has no foreign key because it has no need to link to another table in this design.

Why Transitive Dependencies Are Bad Database Design

What is the value of avoiding transitive dependencies to help ensure 3NF? Let's consider our first table again and see the issues it creates:

AUTHORS

Author_ID	Author	Book	Author_Nationality
Auth_001	Orson Scott Card	Ender's Game	United States
Auth_001	Orson Scott Card	Children of the Mind	United States
Auth_002	Margaret Atwood	The Handmaid's Tale	Canada

This kind of design can contribute to data anomalies and inconsistencies, for example:

If you deleted the two books *Children of the Mind* and *Ender's Game*, you would delete the author "Orson Scott Card" and his nationality completely from the database.

You cannot add a new author to the database unless you also add a book. What if the author is yet unpublished or you don't know the name of a book they authored?

If "Orson Scott Card" changed his citizenship, you would have to change his citizenship in all records in which he appears. Having multiple records with the same author can result in

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You can't delete a book such as *The Handmaid's Tale* without also completely deleting the author.

Frequently Asked Questions

What are the types of dependencies in a database management system? There are four types of database dependencies: trivial functional dependencies, full functional dependencies, transitive dependencies, and multivalued dependencies.

How does functional dependency differ from transitive dependency? In database management, dependency is a relation between two or more attributes (columns). While functional dependency is an association between two attributes of the same relation, transitive dependency occurs when an indirect relationship causes functional dependency.

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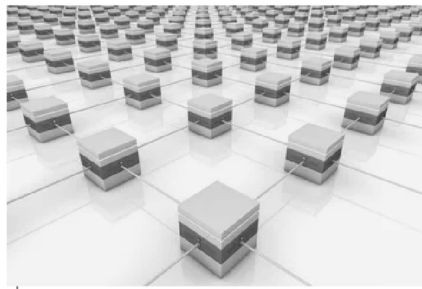


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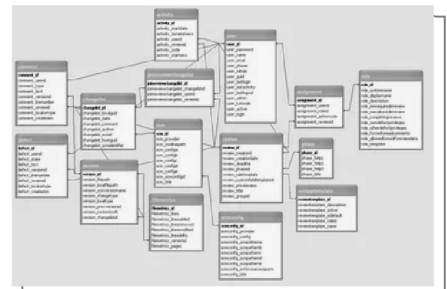
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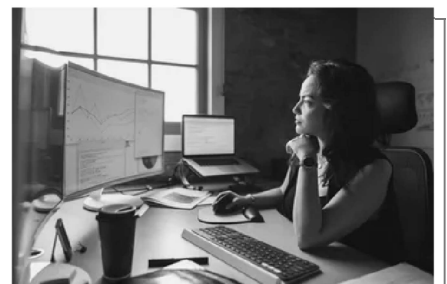
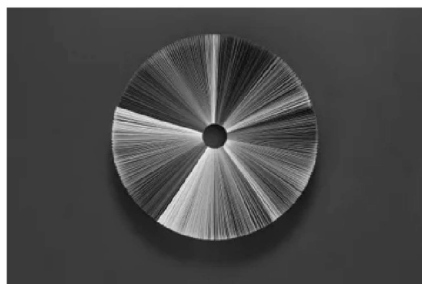
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